

AI FitTrack

Technical Architecture

6.1 Overview

AI FitTrack is designed using a three-tier technical architecture. The architecture separates the user interface, application logic, and database operations into different layers. This makes the system easier to develop, maintain, test, and extend.

6.2 Architecture Layers

1. Presentation Layer

The Presentation Layer provides the interface through which users interact with the AI FitTrack system.

Technologies: HTML, CSS, JavaScript, Bootstrap

Functions: User registration and login; fitness dashboard; BMI input; workout display; diet plan display; progress monitoring.

2. Application Layer

The Application Layer contains the main business logic of the system. PHP is used to process user requests and perform fitness-related calculations.

Technologies: PHP, REST API

Functions: User authentication; BMI calculation; calorie calculation; workout recommendation; diet recommendation; progress tracking; user data processing.

3. Database Layer

The Database Layer stores and manages all information required by the application.

Database: MySQL

Main Tables: User, BMI, Calories, Workout, Diet, Progress

6.3 System Architecture

```
USER / WEB BROWSER
↓
PRESENTATION LAYER – HTML / CSS / JavaScript / Bootstrap
↓
APPLICATION LAYER – PHP Backend / Business Logic
↓
API LAYER – REST API
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DATABASE LAYER – MySQL
```

6.4 Data Flow

The user enters information through the web interface. The request is sent to the PHP application layer. The application processes the request and communicates with the REST API and MySQL database. The processed result is then returned to the user interface.

```
User → Web Interface → PHP Backend → REST API → MySQL Database → PHP Backend → Web Interface → User
```

6.5 Development Environment

Component	Technology
Frontend	HTML, CSS, JavaScript
UI Framework	Bootstrap
Backend	PHP
API	REST API
Database	MySQL
Web Server	Apache
Development Server	XAMPP
Code Editor	Visual Studio Code
Browser	Google Chrome / Microsoft Edge

6.6 Advantages of the Architecture

- Simple and easy to maintain.
- Separates frontend, backend, and database operations.
- Provides better scalability.
- Makes testing easier.
- Allows APIs to be reused by other applications.
- Provides organized management of user fitness data.
- Makes future AI-based fitness recommendations easier to integrate.